

Reading Response:

FamiData Hub: A Speculative Design Exploration with Families on Smart Home Datafication – Wang et al

Summary. Main Takeaways: The paper explores the impacts of datafication on families, specifically how smart homes reduce children's agency and increase (and foster) dependence on technology. It seems as though smart homes exacerbate data risks because of the type of information they collect, such as audio, video, and health records (more sensitive than iPhone or iPad data) that can be used to exploit and monetize certain child behaviors. The study found that when families were presented with comprehensive data collection info (through the "Privacy Practices Panel" in the prototype), they realized they had lost control over family "norms" and were unhappy with the extent of the surveillance. **Strengths/weaknesses and/or limitations?** A key strength is the methodology the researchers adopted, which built upon existing literature (like Zhao et al.) regarding children's awareness of privacy risks and utilized a co-design approach that treated children as equal partners to adults. The study also interestingly employed "speculative design" and "playful by design" frameworks—referencing Bardzell's four dimensions of critical design—by using LEGO not as a toy but as a medium for creative construction to visualize abstract concepts and divergent thinking amongst family members. One limitation is that the recruited families were likely already more informed and worried about these practices than the average population, considering that they opted to participate in the study in the first place. **Questions:** Given that people adopt smart technologies primarily for convenience, why would they want to disrupt that convenience to constantly check data? The proposed solution seems like a band-aid, and I'm not quite convinced that people don't know data is being collected when they buy these products (but I realize this is my bias as a CS major). Also, the "Algorithmic Practices Panel" is somewhat mysterious; how exactly does it demystify the smart system's decision-making for users? Furthermore, there seems to be a paradox where parents want to protect their children's vulnerabilities yet continue to adopt these invasive devices; why does the desire for autonomy not outweigh the adoption? Finally, it seems puzzling that children suggested using *more* technology (like a robot) to solve the problems of data collection. Did the researchers or parents challenge this reliance on tech to fix tech?

Reaction. Most interesting/thought-provoking: It was interesting that smart home data is uniquely difficult to process because it conflates family data with individual data, making it hard to protect children without sacrificing family-wide benefits. I, again wonder if the consequences of smart home devices' data practices outweigh the benefits for parents. Additionally, specific family dynamics were surprising, such as an 8-year-old's cognizance of location tracking and a child's specific fear that a smart fridge might use data to poison them. This, coupled with the tension between children wanting personalized data autonomy (separate from their parents) and the somewhat alarming concept of creating digital twins, definitely made the need for digital literacy surrounding data collection from smart home devices apparent. **Applications:** Future designs should probably prioritize transparency in data collection to ensure that the convenience of a device does not overshadow the consequences of its use. Also, applying the "no expectation" or "no right answer" approach used in the study's interviews can help gather honest and exploratory user feedback. Finally, there is a clear need to embed data literacy into the design itself to help families navigate these complex systems